

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

Chapter 10

COST OF PRODUCTION AND REVENUE ANALYSIS

COST OF PRODUCTION:

A firm or producing unit needs different combinations of factors of production i.e. land, labour, capital and organization to produce various quantities of goods and services. The services of these factors are paid in the form of rent, wages and interest by the entrepreneur. Moreover, amount is also paid for raw material, fuel, power, transportation, publicity, taxes etc. Thus cost of production is the sum total of all above mentioned expenses which a firm has to bear for producing a specific quantity of a commodity. Normal profit is also included in it. Cost function can be expressed as under:-

$$C = X_1 \cdot PX_1 + X_2 \cdot PX_2 + X_3 \cdot PX_3 + \dots + X_n \cdot PX_n$$

Here C = Cost of production

$X_1, X_2, X_3 \dots X_n$ = Units of various factors of production

$PX_1, PX_2, PX_3 \dots PX_n$ = Rewards (prices) paid to units of factors

It must be remembered that the reward of self owned resources / factors are also included in the cost of production which is known as (Implicit Cost)

CLASSIFICATION OF COST OF PRODUCTION

COST OF PRODUCTION OF ANY FIRM MAY BE CLASSIFIED AS UNDER:-

I. TOTAL COST (T C):-

Total cost includes all types of expenses faced by the firm while producing a specific quantity of a commodity. It includes rent, wages, interest, price of machines, raw material, transportation and publicity expenditure, taxes and duty paid by the firm, normal profit etc.

Total cost can be divided into two parts:-

- (a) Fixed Cost (Fc) (b) Variable cost (V C)

Thus $T C = F C + V C$

(A) FIXED COST (F C):-

It is also known as "Indirect Cost" or "Supplementary Cost". It is permanent in nature and remains constant in the short-period. Fixed cost does not vary with the changes in the units of out put. Firm has to bear this cost even if production is zero or factory is closed. It includes price of land, rent of building, price of machines installed, wages of permanent employee's, interest on debt etc.

(B) VARIABLE COST (V C):-

It is also regarded as "Direct Cost" or "Prime Cost". It changes with the change in the

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units of output. The more the units of production the higher the variable cost. Variable cost falls with low level of production and becomes zero at zero production or shut down position of the firm.

Variable cost includes price of raw material, wages to workers, depreciation and maintenance of machines, transportation cost, power and fuel (Gas, electricity, coal etc.) bills, publicity charges, sales tax, excise duty etc.

2. AVERAGE TOTAL COST (ATC OR A C)

It is cost of average per unit output. It is calculated while dividing total cost by total units of output.

$$\text{Thus, Average total cost} = \frac{\text{Total Cost}}{\text{Total Units of Output}}$$

$$\text{ATC or AC} = \frac{\text{TC}}{\text{Q}}$$

If a firm produces 100 units of commodity X per day and its total cost = Rs. 8000.

$$\text{Then AC} = \frac{8000}{100} = 80 \text{ Rs.}$$

Average Cost (AC) can be divided into two parts:-

(a) Average Fixed Cost (AFC):-

When fixed cost is divided by total units of output, we arrive at average fixed cost.

$$\text{AFC} = \frac{\text{Fixed Cost}}{\text{Units of Output}} = \frac{\text{FC}}{\text{Q}}$$

Suppose FC = 2000 while units of output (Q) of commodity X per day = 100

$$\text{Then AFC} = \frac{\text{FC}}{\text{Q}} = \frac{2000}{100} = 20 \text{ Rs.}$$

It must be noted that AFC goes on decreasing with the increase in the units of output of the commodity being produced (the reason is unchanged fixed costs).

(b) Average Variable Cost (AVC):-

In order to achieve AVC, variable cost is divided by the units of output.

$$\text{Thus, AVC} = \frac{\text{VC}}{\text{Q}}$$

For example, VC = 6000 and units of output (Q) = 100

$$\text{Then AVC} = \frac{\text{VC}}{\text{Q}} = \frac{6000}{100} = 60 \text{ Rs.}$$

As variable cost changes with the variation in the units of output. Thus AVC also varies accordingly, declining initially, reaching at its minimum and then increasing again being U-shaped or saucer shaped (—) AVC curve.

3. MARGINAL COST (MC):-

Marginal cost concept is very crucial in Microeconomic theory. Marginal cost (MC) is the extra, or additional cost, of producing one more unit of output. MC can be determined for each additional unit of output by noting the change in total cost which that unit's production entails.

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$$\text{Thus MC} = \frac{\text{Change in TC}}{\text{Change in Q}} = \frac{\Delta \text{TC}}{\Delta Q} = \frac{d\text{TC}}{dQ}$$

$$\text{Suppose Q} = 100 \text{ and TC} = 8000$$

$$\text{Now new Q} = 101 \text{ and TC} = 8075$$

$$\text{Thus } \Delta Q = 1 \text{ and } \Delta \text{TC} = 75$$

$$\text{MC} = \frac{75}{1} \text{ OR } 8075 - 8000 = 75$$

Various concepts of costs can be explained in view of time period.

1. SHORT PERIOD COST:-

Short run is a period of time in which size of the plant cannot be changed. No new entry is possible constant factors being unchanged, fixed cost remains the same. In view of an increase in the demand of output. Production and supply can be increased up to a certain limit. Such a limit depends upon the excess capacity of the plant, changes of an extra shift and availability of the stock of product.

Short period relationship among FC, VC, TC, AFC, AVC, ATC and MC has shown in the following table.

1	2	3	4	5	6	7	8
Units of output (Q)	FC	VC	TC 2+3	AFC	AVC $\frac{3}{1}$	ATC $\frac{4}{1}$	MC
0	20	—	20	—	—	—	—
1	20	60	80	20	60	80	60
2	20	100	120	10	50	60	40
3	20	130	150	6.6	43.3	50	30
4	20	150	170	5	37.5	42.5	20
5	20	160	180	4	32	36	10
6	20	180	200	3.3	30	33.3	20
7	20	210	230	2.9	30	32.9	30
8	20	250	270	2.5	31.2	33.7	40
9	20	300	320	2.2	33.3	35.5	50
10	20	360	380	2	36	38	60

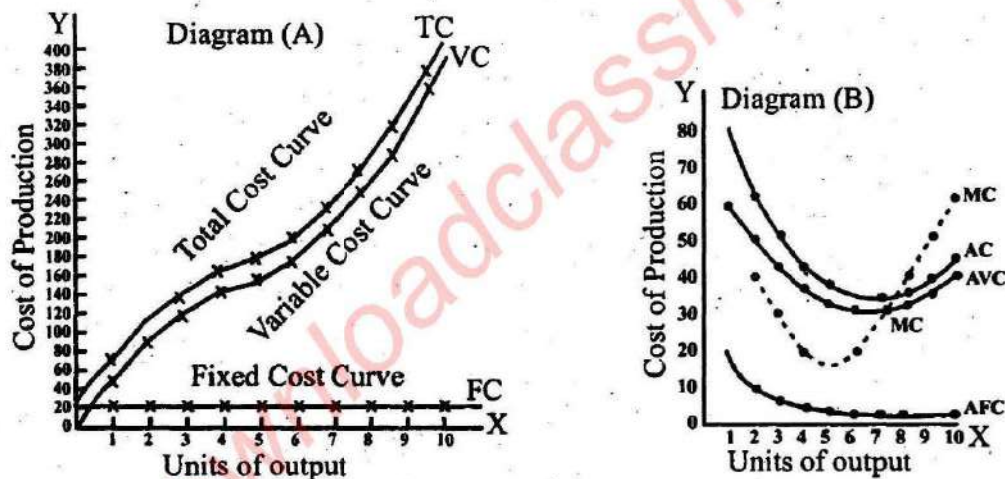
MAIN POINTS OF THE PREVIOUS SCHEDULE:-

1. Column No. 1 shows units of output.
2. Column No. 2 shows fixed cost which remains unchanged from Zero output to the last unit of output.
3. In Column No. 3 variable cost has been shown, which rises at decreasing rate upto 5th

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- units of output, after being minimum, it starts rising at increasing rate.
4. Column No. 4 shows total cost (TC) which is the sum of Column No. 2 and 3. $TC = FC + VC$.
 5. In Column No. 5 average fixed cost (AFC) is calculated by $\frac{FC}{Q} \left(\frac{\text{Column No. 2}}{\text{Column No. 1}} \right)$
 6. Column No. 6 shows average variable cost (AVC) as $\frac{VC}{Q} \left(\frac{\text{Column No. 3}}{\text{Column No. 1}} \right)$
 7. In Column No. 7 ATC is indicated as Column $\frac{4}{1}$.
 8. Column No. 8 shows marginal cost (MC) which has been calculated as under:-
 TC of Column No. 2-1, 3-2, 4-3, , 10-9
 Thus $MC = 80 - 20 = 60$, $120 - 80 = 40$, $150 - 120 = 30$ and so on.

DIAGRAPHIC REPRESENTATION OF SHORT-RUN COST CURVES.



EXPLANATION:-

DIAGRAM (A) SHOWS THE FOLLOWING COST CURVES:-

- (a) Fixed cost curve (FC) is parallel to x-axis which shows that fixed cost of a producing unit does not change with the change in output level.
- (b) Variable cost curve (VC) starts from zero. At this point firm is not functioning and there is no output but after it as the output level goes on increasing variable cost curve (VC) also rises upward to the right.
- (c) Total cost curve (TC) starts from 20 (FC at Zero output) and goes on rising upwardly with the increase in output level.

TC curve remains above VC curve and difference at every level of output is equal to FC (20) as $TC = VC + FC$ or $FC = TC - VC$. Thus TC moves parallel to VC while sloping upward to right.

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DIAGRAM (B) EXHIBITS THE FOLLOWING COST CURVES:-

- (a) AFC is the average fixed cost curve of a producing unit. As $AFC = \frac{FC}{Q}$, Thus FC remaining unchanged an increase in output (Q) results decline in AFC and it's curve is sloped downward to the right.
- (b) Average variable cost curve (AVC) initially starts falling. After touching its minimum level, it rises and becomes closer to the AC curve.
- (c) Average cost curve (AC) also falls touching its minimum level starts rising upwardly with the increase in output level, while falling or rising, it remains above average variable cost curve (AVC).
- (d) Marginal cost curve (MC) also falls at initial stages, then with the further increase in output level it starts moving upward to right. It falls as well as rises more sharply than AC and AVC curves. While falling it remains below these curves but when it intersects AC and AVC curves at their minimum points, it becomes above AVC and AC curves because of a sharp rise in it.

The fall in AC, AVC and MC is the result of the application of the law of increasing returns while the rise in these curves is due to law of decreasing returns. As it is the short-run analysis thus AC, AVC and MC are U-shaped.

LONG-RUN COST CURVES

Long-run is the period of time in which new firms may enter and old firms may leave the industry. Size of plant, building and machinery can also be changed according to the rise and fall of demand.

Fixed cost also varies, hence difference between FC and VC is no more visible in the long-run.

LONG-RUN AVERAGE (LAC) AND MARGINAL (LMC) COST CURVES

In the long-run LAC and LMC also initially fall and then start rising. As the period is long thus these curves while falling or rising are very slow and take more time in every range. These are more flatter and saucer (∩) shaped.

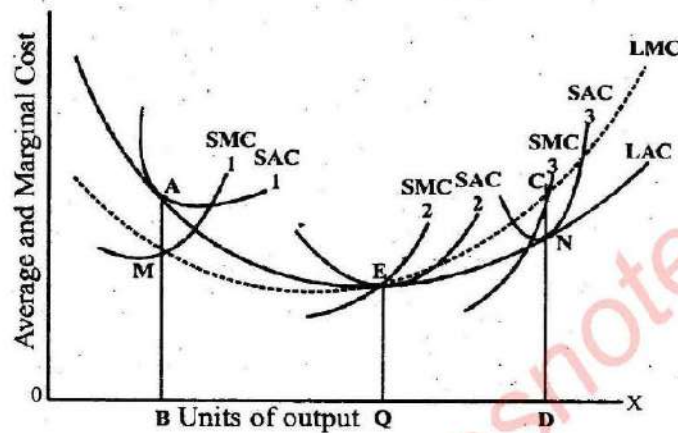
The reasons may be mentioned as under:-

1. Fixed costs do not remain un-changed in the long-run, these costs become variable and fall with the increase in the output level for a longer time and their rise also takes more time, thus LAC becomes more flatter.
2. Due to expansion in producing unit, large scale internal and external economies are available resulting a long-ranged fall in LAC and LMC but when these economies take the shape of dis-economies then LAC and LMC start rising very slowly and become saucer (∩) shaped.
3. In the long-run optimal combination of factors of production is possible. It results a long-ranged fall in LAC and LMC but when such a limit is crossed then LAC and LMC start moving upward to right at a slower rate.

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4. Large scale and long-period production process provides chances for better administration. Thus production process passes through a better position for a longer period but ultimately the application of law of decreasing returns affects this process inversely pushing LAC and LMC upward to right very smoothly.

Following diagram shows the relationship and slope of LAC, LMC, SAC and SMC curves.



1. DERIVATION OF LAC:-

In the diagram we assume that firm produces a commodity with three different sizes of plants. SAC is the short-run average cost curve of Plant No. 1 which is relatively a smaller plant. With permanent increase in demand of the product, Firm installs Plant No. 2 and then Plant No. 3 respectively which are relatively bigger plants in size. Firm's short-run average cost curves now are SAC_2 and SAC_3 . Minimum levels of SAC_1 , SAC_2 and SAC_3 are indicated by points A, E and N respectively where output is OB, OQ and OD while connecting points A, E and N. We arrive at long-run average cost curve (LAC). This curve (LAC) becomes tangent to SAC_1 , SAC_2 and SAC_3 at these points.

LAC is also known as Envelope Curve because of its peculiar shape. The lowest point of LAC is E which indicates the optimum size of the firm.

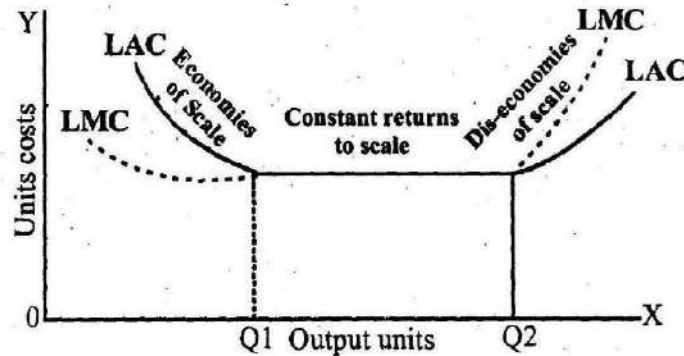
2. DERIVATION OF LMC:-

LMC can be derived from short run marginal cost curves SMC_1 , SMC_2 and SMC_3 as shown in the above diagram. AB the perpendicular curve intersects SMC_1 at point M which lies both on short-run and long-run average cost curves. So M is the point which provides basis for LMC. E is the common point of SAC_2 and SMC_2 which also lies on LAC. Against the common point N on SAC_3 and LAC, C is the point which lies on SMC_3 . Thus by joining M, E and C we arrive at LMC. Although shape of SMC's and LMC is similar yet LMC is flatter than short-run marginal cost curves (SMC_1 , SMC_2 , SMC_3).

MODERN COST OF PRODUCTION (LAC AND LMC)

Modern view regarding LAC and LMC is that where economies of scale are rather rapidly exhausted and dis-economies not encountered until a considerably large scale of output has been achieved, LAC and LMC will remain constant over a wide range of output. This situation has been shown as under:-

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ANALYSIS OF FIRM'S REVENUE

The main aim of every firm is to maximize its profit.

While profit = Total revenue – Total cost

$$\pi = TR - TC$$

Thus revenue concepts must be discussed along with cost concepts to find out the output units of the firm which will bring maximum profit.

There are three main concepts of revenue:-

1. TOTAL REVENUE (TR):-

It is the total amount obtained by a firm after selling its' total output produced during a specific time period. Thus $TR = P \cdot Q$

If a firm produces 100 units of commodity X per day which is sold at Rs. 80 per unit.
 Total revenue = $100 \times 80 = 8000$ Rs.

2. AVERAGE REVENUE (AR):-

Average per unit amount received by a firm is known as AR. It is actually the price received per unit of product. Thus $AR = P$ while $AR = \frac{TR}{Q}$ which means that in order to know AR, TR is divided by units of output sold (Q).

Suppose $TR = 8000$ at $Q = 100$, Thus $AR = \frac{8000}{100} = 80$ Rs.

3. MARGINAL REVENUE (MR):-

It is the net addition in total revenue (TR) after producing and selling an additional unit of output. Suppose, $TR = 8000$ at $Q = 100$ when an additional unit is produced and sold $Q = 101$ and $TR = 8075$. Then $MR = 8075 - 8000 = 75$ (Rs.). Thus MR is the revenue received by selling an additional or last unit of output. MR also indicates the rate of change of TR.

$$\text{Hence } MR = \frac{\Delta TR}{\Delta Q} = \frac{dTR}{dQ}$$

REVENUE CURVES UNDER PERFECT COMPETITION:

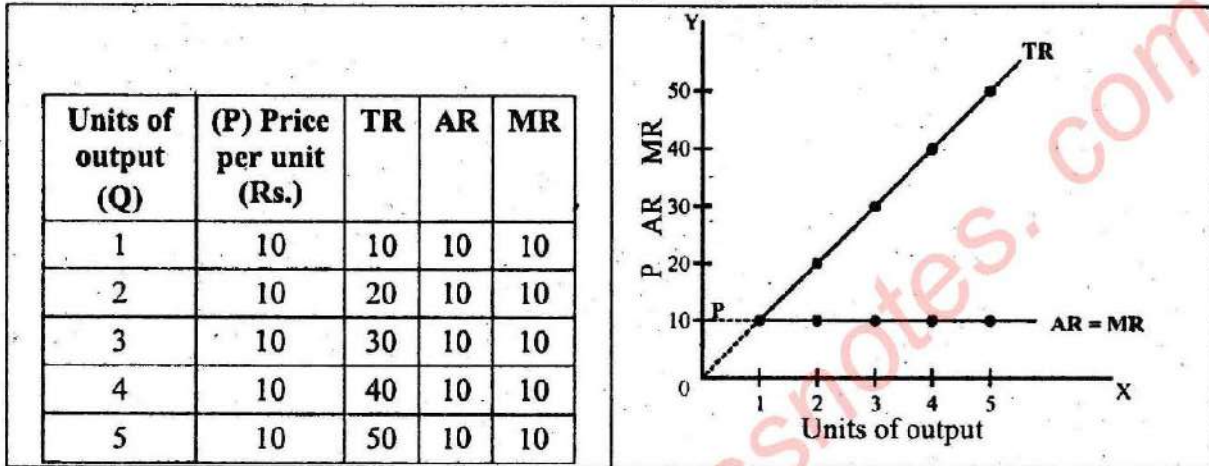
Under perfect competition following conditions are fulfilled in the market.

1. Homogeneous product.
2. Free entry and exit in the market.
3. Large number of buyers and sellers.

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4. Elastic supply of factors of production.
5. Perfect knowledge of the market.

In view of above mentioned conditions no individual buyer or seller can influence market price. Thus price remains the same at all levels of selling. As $P = AR$ thus AR and MR also do not change as shown in the following table and diagram.



Above table shows that price per unit output remains the same (Rs.10) at all sale levels. As $P = AR$ thus average revenue and marginal revenue also remains un-changed. Total revenue goes on increasing as more units of output are sold.

It is clear from the diagram that price (P) average revenue (AR) and marginal (MR) represent a single straight line parallel to x-axis. Thus $P = AR = MR$ under perfect competition but TR curve shows rising trend at a constant rate.

REVENUE CURVE UNDER MONOPOLY AND IMPERFECT COMPETITION:-

Monopoly represents the situation where only one firm has full control over the production / supply of the commodity / service. There is no close substitute available for the product. A monopolist has the power to increase or decrease the price of the commodity. To promote the sale when price is lowered down. Average revenue (AR) follows the change in price and marginal revenue (MR) also tends to change.

As MR falls sharply than AR, Thus while falling MR remains below the AR curve.

Same is the situation under imperfect competition whether it is duopoly, oligopoly or monopolistic competition.

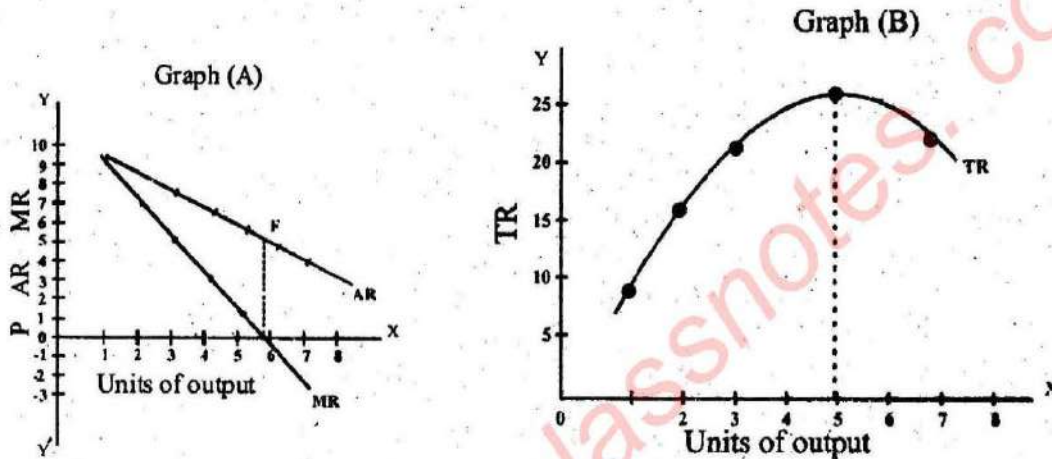
The relationship between AR and MR curves has been shown in the following schedule and graph.

SCHEDULE:

Units of output (Q)	Price (Rs) per unit	TR	AR	MR
1	9	9	9	9
2	8	16	8	7

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3	7	21	7	5
4	6	24	6	3
5	5	25	5	1
6	4	24	4	-1
7	3	21	3	-3



- Graph (A) shows falling AR and MR as with the decrease in price (P) sale of output increases. It must be remembered that:-
 - MR falls sharply and **remains** below the AR curve.
 - MR may become zero or negative but as $AR = P$ thus neither AR is zero nor negative.

- Graph (B) shows increase in total revenue up to output Unit No. 5 where $MR = 1$. After it when MR becomes negative ($MR < 0$) then total revenue start falling. Hence TR rises touches it's maximum height and then falls.

Note: Average revenue curve (AR) is also known as demand curve of the firm. To measure the elasticity of demand at AR, following method is adopted.

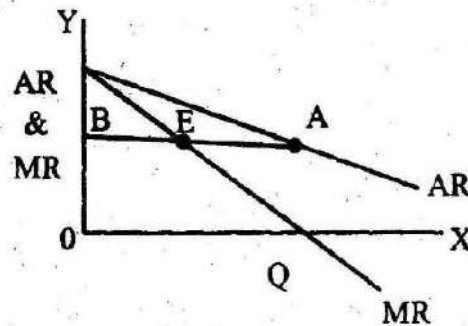
- When $MR = 0$, $E = 1$ [as shown at point F in Graph (A)]
- When $MR > 0$, $E > 1$ [above point F, upper part of AR]
- When $MR < 0$, $E < 1$ [below point F, lower part of AR]

Under Monopoly and Imperfect Competition while falling MR and AR may take following forms. It depends upon the rate of change in price level and TR.

- AR and MR fall in a straight line $AE = BE$ (MR passes through point E)

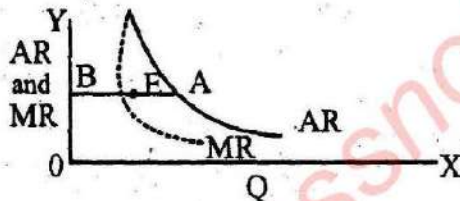
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(a)



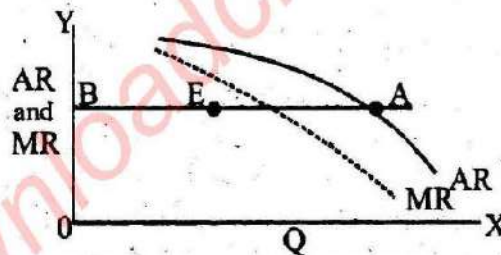
(b) MR moves left of point E when AR and MR are convex to the origin.

(b)



(c) MR moves right to point E when AR and MR are concave to the origin.

(c)



OBJECTIVE QUESTIONS

Select and encircle the correct Answer.

- The right method to find out TR is:
 - Q . P
 - Q . AC
 - TR - TC
 - $\frac{TC}{Q}$
- When firms $TR = TC$ then it faces:
 - No profit and no loss
 - Abnormal profit
 - Loss
 - Normal profit
- Under perfect competition relationship between AR and MR is:
 - $AR < MR$
 - $AR > MR$
 - AR falls, MR rises
 - $AR = MR$
- Which of the following is not included in variable cost (VC)
 - Wages of permanent employees and price of machines
 - Wages of ordinary workers.
 - Transportation cost
 - Publicity Expenses

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5. Short-run AC and MC are:
(a) Flatter (b) -tively sloped (c) U-shaped (d) Elliptical
6. Which is not included in direct or prime cost?
(a) Depreciation of Machines (b) Transportation cost
(c) Salaries of permanent employees (d) Sales tax
7. Short-run relationship between TC and VC shows that:
(a) While falling TC remains under (below) VC curve.
(b) When rising VC shifts upward to TC.
(c) While rising VC remains under TC.
(d) As $TC = AC$ thus a single curve represents both.
8. Which one is an envelop curve.
(a) SAC (b) LAC (c) SMC (d) TC
9. MC shifts upward to VC:
(a) When both are falling (b) When $MC = VC$
(c) After MC intersects VC and moves rising
(d) MC is at its minimum position.
10. Which represents demand curve?:
(a) MR (b) AR (c) TR (d) MC
11. Which is included in variable cost (VC)?:
(a) Rent of the building (b) Price of raw material
(c) Price of machines (d) Wages of permanent employees
12. Which curve intersects the other curves at its minimum level?:
(a) AC to MC (b) FC to TC
(c) MC to AC (d) AVC to ATC
13. $P = AR = MR$, under the situation:
(a) Monopoly (b) Duopoly
(c) Perfect competition (d) Monopolistic competition.
14. In case of monopoly:
(a) MR slopes upward (b) TR always slopes upward
(c) $MR = AR$ (d) $MR < AR$ (MR is always less than AR)
15. When $MR = 0$, elasticity will be:
(a) $E = 0$ (b) $E = 1$ (c) $E > 1$ (d) $E < 1$

DESCRIPTIVE PART - I

SHORT NOTES

1. Differentiate between fixed cost (FC) and Variable Cost (VC)

Fixed cost does not vary with the change in the unit of output. These are permanent in nature. But even in the short period variable cost changes with the change in the units of output.

2. Define marginal Cost (MC) of a firm?

Marginal Cost can be determined for each additional unit of output by noting the change in TC which that unit's production entails $MC = \frac{\Delta TC}{\Delta Q}$

3. How average total cost (ATC or AC) is calculated?

It is cost of average per unit output.

$$ATC = \frac{TC}{Q} = \frac{\text{Total Cost}}{\text{Total units of output}}$$

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4. Define Total Cost (TC)?

All types of expenses faced by a firm such as production cost, selling costs, taxes etc. while producing a specific quantity of a commodity is called TC. It includes fixed cost and variable cost. Thus, $TC = FC + VC$.

5. Why short run average and marginal cost curves are U-Shaped?

Short run AC and MC fall sharply reaches at minimum these costs point then raises more sharply falls due to law of increasing returns and rise due to law of decreasing returns.

6. What is an envelope curve?

LAC is known as an envelope curve because of its peculiar shape and SAC on it. In the long run due to internal and external economies the law of decreasing return may be avoided for a longer time.

7. Bring out difference between AFC and AVC curves?

As FC remains unchanged, and AFC declines with the increase in output and its curve sloped downward to right AVC initially starts falling after minimum level it rises.

8. What is the relationship between AC and MC?

Initially AC and MC fall while decreasing MC remains below AC due to sharp fall, after touching their minimum points both curves start rising and MC goes up to AC due to sharp rise. MC intersects AC at its minimum.

9. Define implicit cost?

Reward of self-owned resources/factors are also included in the cost of production which is known as implicit cost. Such as; the rent of own house/ land, wages of self work, interest of own capital.

10. Define total revenue (TR) of a firm?

It is the total amount obtained by a firm selling its total output produced during a specific time period. Thus $TR = P \cdot Q$. If $p=10$ and $q=100$ Then $TR = 10 \times 100 = 1000$

11. What is meant by average revenue (AR)?

Average per unit amount received by a firm is known as Average Revenue. $AR = P$ we can derive AR while dividing Total Revenue by the quantity of output.

$$\text{Average Revenue} = \frac{TR}{Q}$$

12. What do you know about Marginal Revenue (MR) of a firm?

It is net addition in TR after producing and selling an additional unit of output.

$$MR = \frac{\Delta TR}{\Delta Q} \quad \text{OR} \quad \frac{dTR}{dq}$$

13. Derive AR and MR curves under perfect competition?

AR and MR curves under perfect competition remains same and unchanged. A single curve represents the both AR and MR. it became parallel to X-axis. It is neither rising nor falling.

14. Bring out difference in the slope of AR and MR curves under monopoly?

MR falls sharply and remains below the AR curve. MR may become zero or negative but as $AR = P$ thus AR is neither zero nor negative. It always remains positive.

DESCRIPTIVE PART - II

1. Define cost of production and illustrate the relationship among AC, AVC and MC with the help of graph.
2. Differentiate between "short-run" and "long-run" costs.
3. What is the shape of AR and MR curves under Perfect Competition and Monopoly? Make diagrams.
4. Bring out difference of TR under Perfect Competition and Imperfect Competition.